



AuthiChain Complete Setup Guide

Overview

AuthiChain is a premium NFT authentication and verification marketplace platform with enterprise features, payment processing, and blockchain integration.



Quick Start (3 Minutes)

Option 1: Automated Setup (Recommended)

```
cd /home/ubuntu/authichain_premium
./setup.sh
```

Option 2: Manual Setup

```
cd /home/ubuntu/authichain_premium/app
npm install
npx prisma generate
npx prisma db push
npm run build
npm run dev
```



Prerequisites

Required Software

- ☒ Node.js 18+ (Already installed)
- ☒ npm 9+ (Already installed)
- ☒ PostgreSQL Database (Already configured)

Optional Tools

- Git (for version control)
 - Chrome/Firefox (for testing)
 - Stripe Account (for payments)
 - Alchemy Account (for blockchain)
-



Configuration

1. Environment Variables (.env)

The `.env` file is already configured with test credentials:

```
# Database
DATABASE_URL="postgresql://..."

# Authentication
NEXTAUTH_SECRET="TMZDqgBKn0PbfffMNZvdxGGblN7Pdvw"
NEXTAUTH_URL="http://localhost:3000"

# Stripe (TEST mode - safe for development)
STRIPE_SECRET_KEY="sk_test_..."
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY="pk_test_..."
STRIPE_PRO_PRICE_ID="price_..."
STRIPE_BRAND_PRICE_ID="price_..."
STRIPE_WEBHOOK_SECRET="whsec_..."
```

2. Stripe Configuration (For Live Payments)

Get Your Stripe Keys:

1. Go to [Stripe Dashboard](https://dashboard.stripe.com) (https://dashboard.stripe.com)
2. Navigate to **Developers → API Keys**
3. Copy your keys:
 - Publishable key (starts with `pk_live_`)
 - Secret key (starts with `sk_live_`)

Create Subscription Products:

1. Go to **Products** in Stripe Dashboard
2. Create "AuthiChain Pro" - \$29/month
 - Copy the Price ID (starts with `price_`)
 - Update `STRIPE_PRO_PRICE_ID` in `.env`
3. Create "AuthiChain Brand" - \$99/month
 - Copy the Price ID
 - Update `STRIPE_BRAND_PRICE_ID` in `.env`

Set Up Webhooks:

1. Go to **Developers → Webhooks**
2. Click "Add endpoint"
3. URL: `https://your-domain.com/api/stripe-webhook`
4. Select events:
 - `checkout.session.completed`
 - `customer.subscription.updated`
 - `customer.subscription.deleted`
 - `invoice.payment_succeeded`
 - `invoice.payment_failed`
5. Copy webhook secret (starts with `whsec_`)
6. Update `STRIPE_WEBHOOK_SECRET` in `.env`

3. Database Setup

The database is automatically configured and seeded with:

- ☒ Admin account: `admin@authichain.com` / `Admin123!`
- ☒ Test user: `user@authichain.com` / `User123!`
- ☒ Sample NFTs and collections
- ☒ Sample transactions

Running the Application

Development Mode (with hot reload)

```
cd /home/ubuntu/authichain_premium/app
npm run dev
```

Access at: <http://localhost:3000>

Production Mode

```
cd /home/ubuntu/authichain_premium/app
npm run build
npm run start
```

Background Mode (persistent)

```
cd /home/ubuntu/authichain_premium/app
npm run dev > /dev/null 2>&1 &
```

Testing

Test Stripe Payments (Test Mode)

Use these test card numbers:

- **Success:** 4242 4242 4242 4242
- **Decline:** 4000 0000 0000 0002
- **3D Secure:** 4000 0025 0000 3155

Any future expiry date and any 3-digit CVC will work.

Test Accounts

Role	Email	Password	Features
Admin	ad-min@authichain.com	Admin123!	Full access, analytics, user management
User	user@authichain.com	User123!	Standard user, can buy/sell NFTs

Test Workflow

1. Sign in with test account
2. Browse NFT marketplace
3. Connect wallet (MetaMask recommended)
4. Purchase NFT with test card
5. Verify transaction in admin dashboard



Features



Core Features

- **NFT Marketplace:** Browse, buy, sell authentic NFTs
- **Authentication System:** Secure blockchain verification
- **Payment Processing:** Stripe, PayPal, Crypto payments
- **Subscription Plans:** Pro (\$29/mo) and Brand (\$99/mo)
- **Admin Dashboard:** Analytics, user management
- **Mobile PWA:** Progressive Web App support
- **API Integration:** RESTful API with documentation



Enterprise Features

- **White-label Solutions:** Customizable branding
- **Multi-currency Support:** USD, EUR, GBP, JPY, etc.
- **Escrow System:** Secure NFT transactions
- **Compliance Tools:** KYC/AML integration ready
- **Analytics Dashboard:** Real-time metrics
- **Role-based Access:** Granular permissions



Deployment Options

Option 1: Vercel (Recommended)

```
npm install -g vercel
vercel login
vercel deploy
```

Option 2: Docker

```
docker build -t authichain .
docker run -p 3000:3000 authichain
```

Option 3: Traditional VPS

```
# Install PM2
npm install -g pm2

# Start application
pm2 start npm --name "authichain" -- start

# Set up auto-restart
pm2 startup
pm2 save
```



Security Checklist

Before Going Live:

- [] Replace all test Stripe keys with live keys
- [] Set strong `NEXTAUTH_SECRET` (32+ characters)
- [] Enable HTTPS/SSL certificate
- [] Configure CORS properly
- [] Set up rate limiting
- [] Enable webhook signature verification
- [] Configure backup system for database
- [] Set up monitoring and alerts
- [] Review and test all payment flows
- [] Implement logging and error tracking



Project Structure

```

authichain_premium/
├── app/
│   ├── api/           # API routes
│   ├── app/           # Next.js 13+ app directory
│   ├── components/    # React components
│   ├── hooks/         # Custom React hooks
│   ├── lib/           # Utility libraries
│   ├── prisma/        # Database schema
│   ├── public/        # Static assets
│   ├── scripts/       # Setup and utility scripts
│   ├── types/         # TypeScript types
│   ├── .env           # Environment variables
│   ├── package.json   # Dependencies
│   └── tsconfig.json  # TypeScript config
├── setup.sh           # Automated setup script
└── SETUP_GUIDE.md    # This file

```



Troubleshooting

Issue: Port 3000 already in use

```

# Find and kill process
lsof -ti:3000 | xargs kill -9

# Or use different port
PORT=3001 npm run dev

```

Issue: Database connection failed

```
# Check database is running
npx prisma db push

# Reset database
npx prisma migrate reset
```

Issue: Stripe webhook not receiving events

1. Check webhook URL is correct
2. Verify webhook secret in .env
3. Test with Stripe CLI:

```
stripe listen --forward-to localhost:3000/api/stripe-webhook
```

Issue: Build errors

```
# Clear cache and rebuild
rm -rf .next node_modules
npm install
npm run build
```

Support & Resources

Documentation

- [Next.js Docs](https://nextjs.org/docs) (https://nextjs.org/docs)
- [Stripe Docs](https://stripe.com/docs) (https://stripe.com/docs)
- [Prisma Docs](https://www.prisma.io/docs) (https://www.prisma.io/docs)

Getting Help

- Check existing documentation
- Review error logs: `.logs/` directory
- Test in development mode first
- Verify environment variables



Success Metrics

After setup, you should be able to:

- ☒ Access application at `http://localhost:3000`
- ☒ Sign in with test accounts
- ☒ Browse NFT marketplace
- ☒ Process test payments
- ☒ View admin dashboard
- ☒ Test API endpoints



Next Steps

1. **Customize Branding:** Update logos, colors, copy
 2. **Configure Payments:** Set up live Stripe account
 3. **Add Content:** Upload NFTs and collections
 4. **Test Everything:** Run through all user flows
 5. **Deploy to Production:** Choose hosting provider
 6. **Monitor & Iterate:** Track analytics, gather feedback
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Ready to Launch!

Your AuthiChain platform is now set up and ready to authenticate and verify NFTs at scale!

For the latest updates and features, check the main repository.

Version: 1.0.0

Last Updated: October 2025

Platform: AuthiChain NFT Marketplace